KOR-ACC-29 Huawei Access Node Hardware/Environmental Fault Resolution SOP

Node: KOR-ACC-29 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_6626

Description

In the Huawei Access Node, the Line Card (LC) is a critical component responsible for handling high-speed data transmission over optical fibers. The Line Card Crash (LCC) failure domain can occur due to ASIC memory parity errors, which cause the LC to generate a Major Alarm (ALARM_6626) indicating a hardware or environmental fault. This failure typically results from a malfunctioning Line Card, which can be attributed to various root causes such as hardware defects, software bugs, or environmental issues like temperature or power supply fluctuations. The loss of redundancy in the power supply system can also trigger this failure, as the Line Card relies on a redundant power supply to ensure continuous operation. To mitigate this failure, it is crucial to identify the root cause and perform the necessary repairs or replacements to prevent future occurrences.

Pre-requisite Checks

- Verify that the Koramangala Datacenter is accessible and that the necessary personnel are present to perform the required tasks.

- Check the redundant link status to ensure that it is functioning correctly and that the data transmission is not affected.

- Gather information about the failure from the Huawei Access Node's Alarm Log and System Log.

Potential Root Causes

Cause	Probability
Hardware Defect	High
Software Bug	Medium
Environmental Issue (Temperature/Power Supply Fluctuations)	Low
Loss of Redundancy in Power Supply System	High

Diagnostic Commands & Expected Outputs

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Command: display alarm-alarmlog, Expected Output: ALARM_6626: Major Alarm, Line Card Crash, Reason: ASIC memory parity error, Description: This command displays the Alarm Log and provides information about the most recent alarm event. The output indicates the severity and reason for the alarm, which in this case is a Major Alarm due to a Line Card Crash caused by ASIC memory parity errors.

Command: display hardware, Expected Output: Line Card (LC) Status: Down, Reason: Hardware Failure, Description: This command displays the
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current status of the hardware components, including the Line Card. The output indicates that the Line Card is Down due to a hardware failure.

Command: display power supply, Expected Output: Power Supply Status: Redundant Power Supply Lost, Description: This command displays the status of the power supply system. The output indicates that the redundant power supply is lost, which can cause the Line Card to crash.
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Resolution Steps

Step 1: Step: Replace the faulty Line Card, Command: replace line-card 0/0/0, Description: Replace the faulty Line Card with a new one to ensure continuous operation of the network. Perform this step carefully, following the manufacturer's instructions and guidelines to prevent any damage to the equipment or data transmission disruption.

Step 2: Step: Restore the redundant power supply system, Command: power supply enable 1, Description: Restore the redundant power supply system to ensure that the Line Card has a backup power source in case of a failure. Perform this step carefully, following the manufacturer's instructions and guidelines to prevent any damage to the equipment or data transmission disruption.

Post-Check Verification

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Command: display alarm-alarmlog, Expected Output: No alarm events, Description: Verify that the alarm event has been cleared by checking the Alarm Log. If the output indicates no alarm events, it means that the resolution step has been successful.

Command: display hardware, Expected Output: Line Card (LC) Status: Up, Description: Verify that the Line Card is functioning correctly by checking its status. If the output indicates that the Line Card is Up, it means that the resolution step has been successful.
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Roll-back Procedures

- Step: Restore the original Line Card, Command: replace line-card 0/0/0 original, Description: Restore the original Line Card if the replacement Line Card does not resolve the issue. Perform this step carefully, following the manufacturer's instructions and guidelines to prevent any damage to the equipment or data transmission disruption.

- Step: Disable the redundant power supply system, Command: power supply disable 1, Description: Disable the redundant power supply system if it was previously enabled. Perform this step carefully, following the manufacturer's instructions and guidelines to prevent any damage to the equipment or data transmission disruption.